HER INSPIRATION

"For me, people are role models. Everyone has something to share with their wisdom and lived experiences. I just soak that in"



INFOSYS SCIENCE FOUNDATION

DR ANJANA BADRINARAYANAN, 40
Molecular biologist, National Centre for Biological Sciences (NCBS)
Bengaluru

The Cell Mender

An average human cell can sustain up to 10,000 damages to its DNA every day. And these can get fixed pretty fast too. These repair mechanisms are crucial because several diseases, including cancer, are linked to incorrectly fixed cell damage.

Repairing a double strand DNA break, one such lethal damage, requires a matching piece of DNA from elsewhere in the cell. Easier said than done, given how efficiently DNA is packed into a tiny cell. Stretched out end to end, human DNA from a single cell can extend up to 6 feet while that of an invisible bacterium is little more than 1 mm. How then, do pieces of DNA find each other in the crowd? Over the past decade, Anjana Badrinarayanan's lab at the National Centre for Biological Sciences (NCBS) found a missing piece of that puzzle by observing bacterial cells in real time through high-resolution microscopy, much like "ecologists observing animals in their natural habitat". The repair mechanism, they discovered, is a highly regulated process of sweep-search involving specific enzymes that act like a motor. "It grabs on to the piece of DNA and the protein that's doing the DNA search and sends it off on these sweeps. We found the motor that powers this process, giving us insights into the rules governing what is the right match and what is not," explains Badrinarayanan.

WHAT MAKES HER A SHERO

Anjana's groundbreaking research into the molecular mechanisms that help cells repair DNA damage

Winner of the 2025 Infosys prize in Life Sciences, Badrinarayanan's fascination with DNA maintenance in microbes grew during her graduate studies in Oxford. "Microbes are the most diverse living systems on our planet...and we still know remarkably little about how they survive," she says. "Everything from recycling and scavenging resources to driving essential biogeochemical cycles that sustain us depends on microbial life." ■

—Ajay Sukumaran